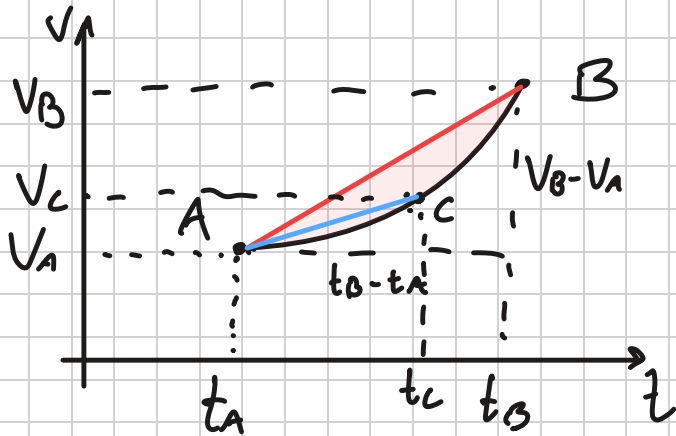


Week 2

instantaneous acceleration

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} \quad \text{where } v(t)$$



on AB :

$$\underline{a_{ave}} = \frac{v_B - v_A}{t_B - t_A}$$

On AC : $a_{ave} = \frac{v_C - v_A}{t_C - t_A}$

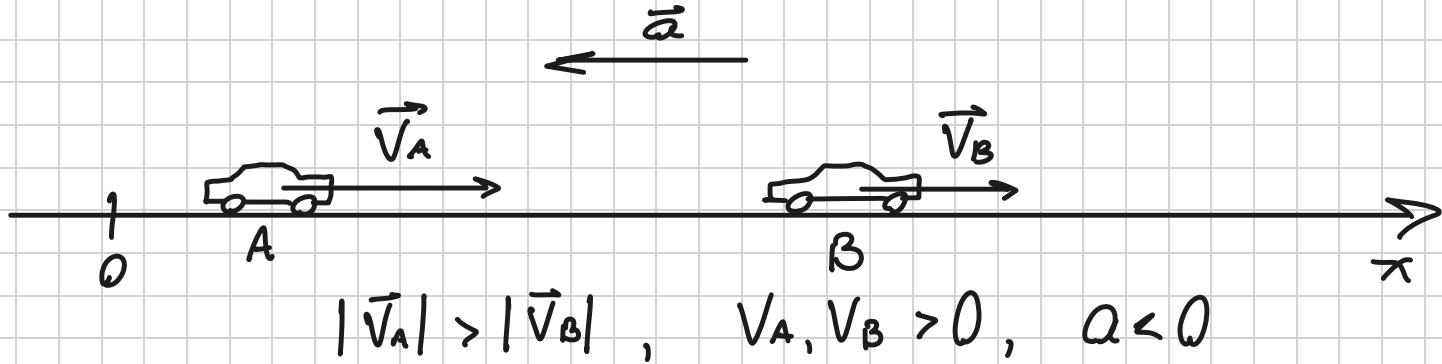
$$a_{ave AB} \neq a_{ave AC}$$

Let C slide toward A such that $t_C - t_A \rightarrow 0$

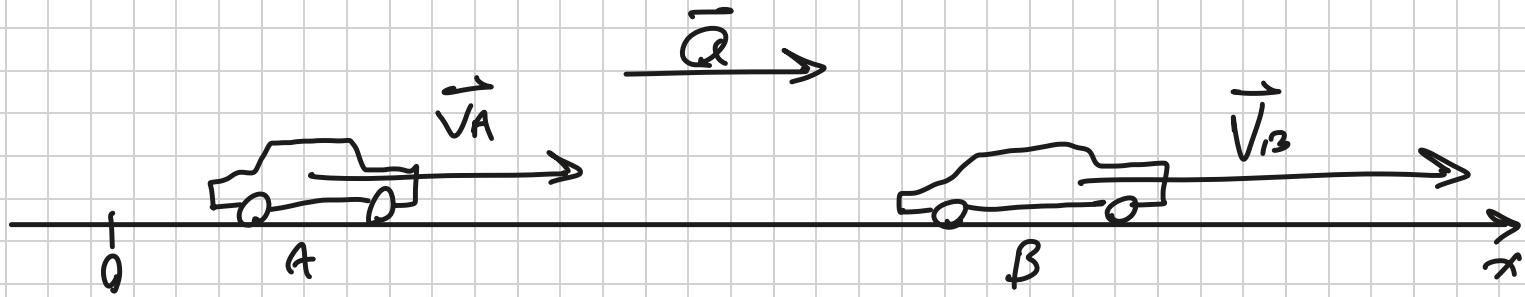
$$a_{inst} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$$

Note : a, v do not necessarily point in the same direction

1.

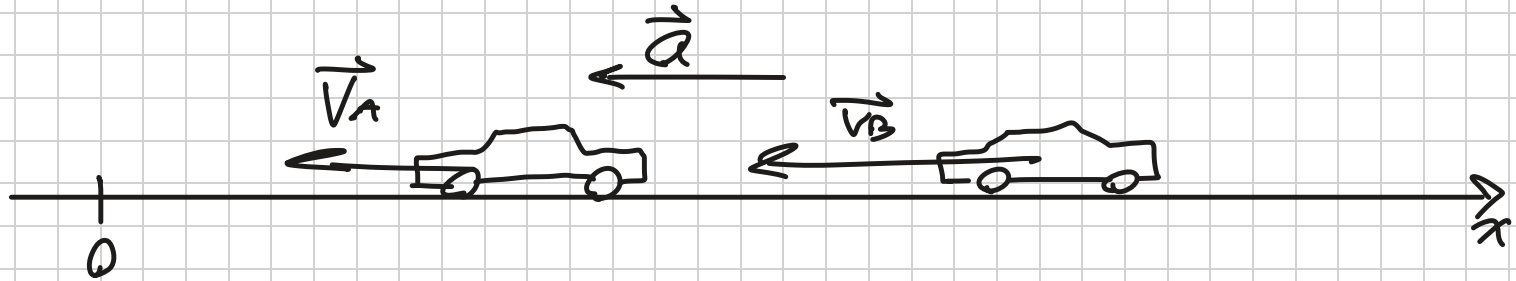


2.

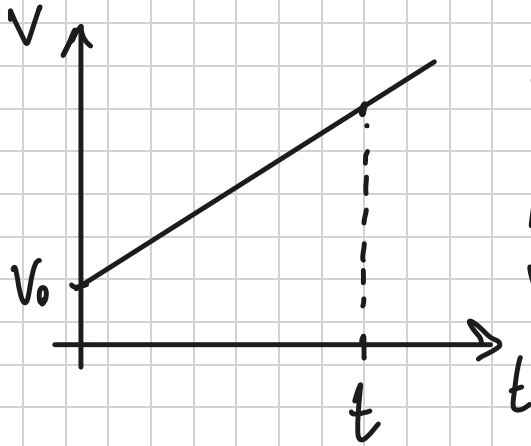


$$|\vec{v}_A| < |\vec{v}_B|, v_A, v_B > 0, Q > 0$$

Exercise: redup diagram. $\vec{v} < 0$



2.4 motion with $\vec{a}$ constant



$$v(t) = c + d \cdot t$$

$$|| v(0) = v_0$$

$$v(0) = c \Rightarrow c = v_0$$

$$\frac{dv}{dt} = d = a$$

$$\Rightarrow \boxed{v(t) = v_0 + at}$$

What is V_{ave} between D and t

$$x(0) = x_0 \quad V_{ave} = \frac{x - x_0}{t}$$

$$\Rightarrow x - x_0 = V_{ave} \cdot t$$

Because $V(t)$ is linear, $V_{ave} = \frac{V_0 + V(t)}{2}$

$$\Rightarrow V_{ave} = \frac{V_0 + V_0 + at}{2} = V_0 + \frac{at}{2}$$

$$\Rightarrow x - x_0 = \left(V_0 + \frac{at}{2}\right)t = V_0 t + \frac{at^2}{2}$$

$$\Rightarrow \boxed{x(t) = x_0 + V_0 t + \frac{1}{2}at^2}$$

$$\frac{dx}{dt} = \frac{d}{dt} \left(x_0 + V_0 t + \frac{1}{2}at^2 \right) = V_0 + at$$

Eliminate t

$$at = V - V_0 \Rightarrow t = \frac{V - V_0}{a}$$

$$\left[x = x_0 + V_0 \frac{V - V_0}{a} + \frac{1}{2} a \left(\frac{V - V_0}{a} \right)^2 \right] \cdot 2a$$

$$2a(x - x_0) = 2V_0(V - V_0) + (V - V_0)^2$$

$$2a(x - x_0) = \cancel{2V_0V} - 2V_0^2 + V^2 + V_0^2 - \cancel{2V_0V}$$

$$2a(x - x_0) = V^2 - V_0^2$$

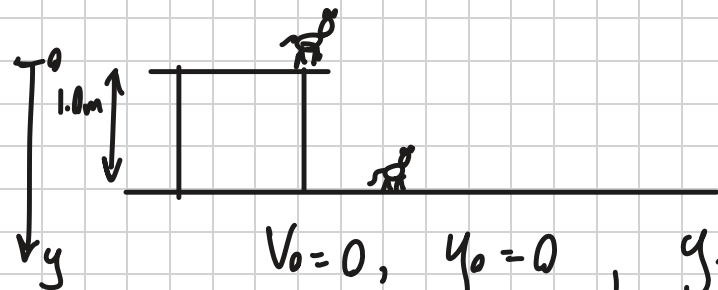
$$V^2 - V_0^2 = 2a \cdot (x - x_0)$$

Let $x - x_0 = d$

$$V^2 - V_0^2 = 2a \cdot d$$

example

Uniform downward $\vec{a} = 9.8 \text{ m/s}^2$



$$V_0 = 0, y_0 = 0, y_1 = 1.0 \text{ m}$$

what's the v of cat when it hits the floor

$$v^2 - V_0^2 = 2g \cdot d \Rightarrow v = \sqrt{2gd}$$

$$v = \sqrt{2 \times 9.8 \times 1.0} = \underline{4.4} \text{ m/s}$$

free fall

- gravity $\vec{a}$ is constant

if air resistance and buoyancy are neglected
 $\Rightarrow$ All objects fall the same acceleration of gravity

if we are close to the earth surface $\Rightarrow$ constant acceleration

$$g = 9.8 \text{ m/s}^2$$


$$a = -g \quad (\text{negative})$$

the 3 basic equations:

$$y = y_0 + v_0 t + \frac{1}{2} a t^2$$

$$y(0) = y_0, \quad v(0) = v_0$$

$$a = -g$$

$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$$

$$v(t) = v_0 + a t$$

$$v(t) = v_0 - g t$$

$$V^2 - V_0^2 = 2a(y - y_0)$$

$$\Rightarrow V^2 = V_0^2 - 2g(y - y_0)$$